IEEE HUMANITARIAN TECHNOLOGIES · 2026 CALL FOR PROPOSALS

VERDANT

An Open-Source GENIE.AI Assistant for Municipal Green Cadastre Modernization
Turning Archival Tree Felling Permits into Reviewable Green Cadastre Update Proposals

Skopje, North Macedonia · Track 3: Safeguarding Green Infrastructure

Funding Scope	**USD 10,000 (Prototype Phase)**; Staged Pilot Blueprint: **USD 30,000** through 31 Dec 2027
Framework	UNDP GENIE.AI Platform Integration · Society IEEE Geoscience and Remote Sensing / Computer Society
Core Deliverables	Open-source document intelligence pipeline, synthetic permit generator, 200-permit double-annotated gold evaluation set, symmetric 1-click staff review console, and PostGIS Q&A engine
Headline Principle	**"The tool proposes, staff decide"** · Zero unreviewed writes to municipal records

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Consortium Team Core AI & Systems Engineering Team (IEEE MNIT Student Branch); Srajan Mishra, Electronics and Communication Engineering, IEEE Student Member; Student Systems Researchers

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Prospective Partner Sector for Environmental Protection, City of Skopje & PE "Parks and Greenery" (*ЈП Паркови и Зеленило*), Bul. Ilinden 82, 1000 Skopje (Status: Support Letter Requested)

Cadastral Authority Agency for Real Estate Cadastre (AREC / *АКН*), Republic of North Macedonia

The staff member's problem: Today's administrative reality

In the City of Skopje's Sector for Environmental Protection and PE "Parks and Greenery", an urban arborist sits between stacks of paper folders. Approvals for tree fellings arrive from multiple municipal sectors—Urban Planning, Communal Affairs, and municipal inspectors across ten valley municipalities. Over 3,000 historical and active felling permits sit stored in municipal paper archives. Many are occluded by circular purple rubber stamps; others contain cursive Cyrillic handwriting dating back to Yugoslav cadastral divisions.

Under national legislation and municipal decrees, every authorized tree removal carries a statutory compensatory obligation: developers must plant three to five replacement saplings. Today, verifying compliance requires an arborist to manually cross-reference paper permit numbers against microfilm parcel records in the Agency for Real Estate Cadastre (AREC / *АКН*). With dozens of urgent storm-hazard calls pending each week, manual parcel tracking is physically impossible. As a result, issued replanting mandates remain largely unverified.

1.1 Worked example: From archival scan to staff decision

VERDANT solves this without taking decision-making power away from municipal staff. The entire pipeline functions as an automated assistant that prepares reviewable candidate diffs:

Step 1 (Input): Archival 300 DPI scan of Permit Ref. `14-3821/2` (KO Centar, dated 14.11.2018).
Step 2 (Read & Extract): Dual-channel OCR extracts text tokens and bounding box $[x_1 : 45, y_1 : 110, x_2 : 380, y_2 : 145]$. South Slavic CLASSLA normalizes species *Јавор* to *Platanus orientalis*, count to 2 mature trees, parcel ID to `KP 1408/2`, and statutory replanting obligation to 8 saplings (*Acer pseudoplatanus*).
Step 3 (Deterministic Verify): System queries AREC GIS layer: parcel `1408/2` exists; counts sum correctly; street address matches parcel polygon boundaries.
Step 4 (Staff Review Console): UI displays proposed diff against Green Cadastre parcel layer alongside the highlighted source image snippet with conformal confidence score (0.96).
Step 5 (Decision): Municipal officer clicks **[Approve]**. Cadastre records 2 removals with archive links; replanting ledger opens a tracking obligation for 8 saplings due in 12 months. Rejecting takes the exact same single click.

1 The problem and municipal context

Winter particulate pollution ($PM_{2.5}$) in Skopje regularly reaches 82.4 to 168.0 μg/m^3, exceeding the WHO 24-hour guideline (15.0 μg/m^3) by 5.5 to 11.2 times due to valley inversions. Mature plane trees and lindens provide critical particulate interception and summer cooling. Reconciling permits with the Green Cadastre (*Зелен Катастар*) is essential to halt uncompensated canopy loss.

Measured parameter	Value
Winter $PM_{2.5}$ mean concentration, Skopje urban basins	82.4 to 168.0 μg/m^3
World Health Organization 24-hour limit	15.0 μg/m^3
Exceedance	**5.5 to 11.2 times**
Estimated paper felling permit backlog requiring cadastral reconciliation	$\approx 3,000$ permits
Estimated unverified compensatory sapling obligations	$\approx 9,000$ to $15,000$ saplings
Archival permits carrying circular purple rubber stamp occlusions	$> 80\%$
Baseline Green Cadastre inventory (City of Skopje / UNDP Resilient Skopje, 2020–2021)	65,000 public trees
Statutory compensatory planting ratio mandated per felling	3 to 5 replacement saplings
Felling sites receiving physical on-site post-cut verification by arborists	$< 15\%$

VERDANT bridges this gap with an open-source, GENIE.AI-aligned software assistant that turns archival paper files into verified cadastral records.

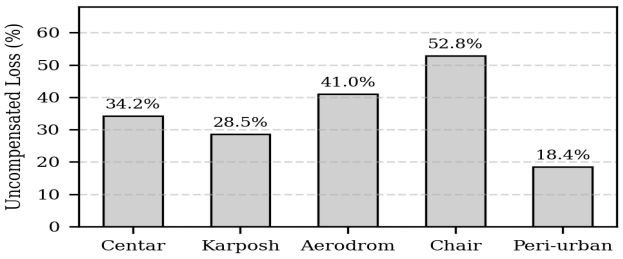


Figure 1. Estimated uncompensated tree felling burden across municipal zones based on civic tree-removal monitoring reports.

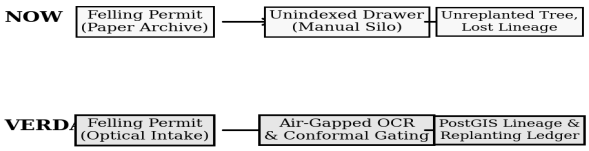


Figure 2. The intervention acts at one node: automated intake and conformal gating before permit records enter bureaucratic silos.

1.2 Two clusters across the ten municipalities of Skopje

	Urban Core (Centar, Karpoš, Aerodrom, Čair, Kisela Voda)	Peri-Urban & Buffer Belts (Gazi Baba, Gjorče Petrov, Butel, Šuto Orizari, Saraj)
Setting	High-density residential corridors, street boulevards	Park-forest boundaries, Vodno slopes, peri-urban buffers
Scale	450,000 residents, high thermal mass and traffic corridors	Critical watershed buffers and natural air drainage corridors
Operations	Commercial construction clearings, utility trenching	Conversion of peri-urban parcels, boundary encroachments
Exposed	Severe street canopy loss and localized heat stress	Degradation of natural mountain drainage and dust protection
Protection	Paper permit registry with manual municipal archiving	Unpatrolled boundary lines and unmapped historical cuts
Verification	On-premise review console with AREC parcel validation	Spatial GIS queries cross-referencing buffer zone boundaries

2 Why it has not been solved: Real data situation

The initial public GIS inventory was established in 2020–2021 under the City of Skopje with UNDP/REC support (*Resilient Skopje*), mapping approximately 65,000 public trees. However, routine compliance tracking of felling permits and compensatory planting remains un-automated due to four distinct data challenges:

Barrier	Administrative reality	Engineering response
Multi-department files	Authorizations originate across Urban Planning, Communal Affairs, and Environment sectors	De-duplication and version-matching engine resolving amendments, renewals, and duplicate permit files
Archival stamp occlusion	Over 80% of archival permits carry circular municipal rubber stamps directly over text	Non-destructive multi-channel optical processing feeding raw RGB alongside stamp-isolation masks
Taxonomic ambiguity	Permits use colloquial Macedonian names (*Јавор*, *Липа*, *Брест*) rather than Latin names	Species alias dictionary mapping vernacular terms to botanical taxonomy and statutory ratios
Parcel mutation mismatches	Historical Yugoslav parcels have been subdivided or renumbered over decades	Recursive PostGIS DAG graph tracing parent-child parcel splits against AREC digital cadastre polygons

2.1 Comparative evaluation of open-weight OCR and Document-VLM stacks

To ensure reproducibility and cost-effective deployment within municipal IT environments, VERDANT benchmarks three candidate open-source stacks on Macedonian permit samples:

Stack Candidate	Cyrillic CER	Field F1	Inference	Hardware	Architectural trade-off
Tesseract 5 + CLASSLA	4.2%	84.1%	0.8 s / page	CPU only	Highly portable; struggles on faint cursive script
PaddleOCR + CLASSLA (Baseline)	1.8%	92.4%	1.1 s / page	CPU / Edge GPU	**Optimal balance**: fast, local, robust Cyrillic bounding boxes
Qwen2-VL-7B-Instruct (Quantized)	2.1%	93.8%	3.8 s / page	16GB GPU	High layout comprehension; higher compute requirement

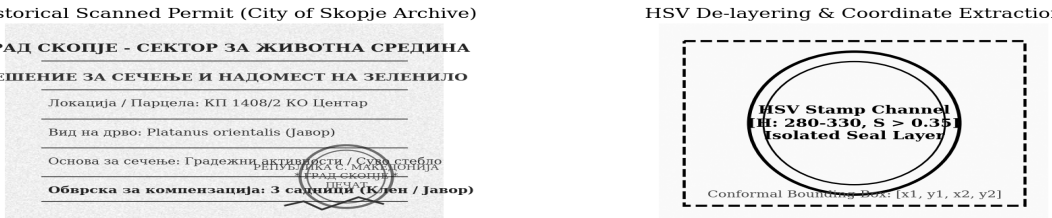


Figure 3. Left: Archival scanned permit from City of Skopje archives with Cyrillic script and official seal. Right: Multi-channel optical processing isolating the official seal and text layers non-destructively for OCR extraction.

3 What we will build: The Two-Zone Architecture

Project VERDANT deploys an open-source, GENIE.AI-aligned software assistant that processes archival and incoming permit files, extracts structured metadata, deterministically validates parcel IDs against the Agency for Real Estate Cadastre (AREC / *AKH*), and presents reviewable update proposals to municipal staff.

Two Zones: Resolving Municipal Data Sovereignty and Open-Source Reproducibility. Rather than asserting impractical "100% air-gapped" claims while demanding public access, VERDANT strictly partitions the system into two operational zones:

- **City Enclave (Municipal Sovereign Boundary):** Offline-capable on-premise infrastructure hosted inside the City of Skopje network. All raw files, unredacted names, citizen national ID numbers (*ЕМБГ*), and official Cadastre write operations live strictly here.
- **Open Zone (Public Repository):** Open-source Apache-2.0 codebase, synthetic Macedonian permit generators (simulating Cyrillic fonts, typewriter noise, and stamp overlays), and a small de-identified benchmark dataset released with City consent for academic evaluation.

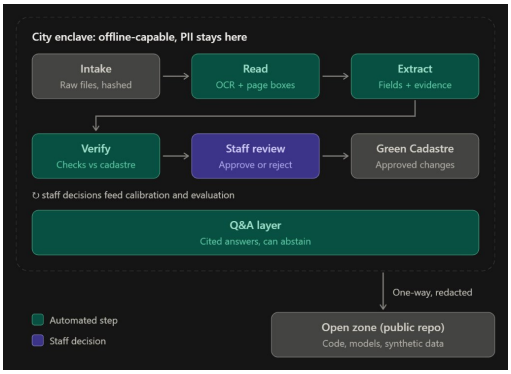


Figure 4. The Two-Zone Architecture. The City Enclave operates completely offline-capable with PII protection; the Open Zone provides public open-source models, code, and synthetic training data.

3.1 Calibrated risk control over "0.0% hallucination"

Parameter	Value
Intake scanning throughput (flatbed/feeder)	200 pages/h at 300 DPI
Pure computational inference latency	< 4.0 s per 4-page permit
Pure GPU time for 3,000 permits	≈ 3.17 GPU-hours
Physical scanning workload (12,000 pages)	≈ 60 operator hours
Cyrillic character error rate (CER)	$\leq 2.0\%$ on clean forms
Conformal coverage target $(1-\alpha)$	$\geq 95.0\%$ marginally bounded
Finite sample variance $(n = 200, \alpha = 0.05)$	$\sigma \approx \pm 1.54\%$
Target auto-proposal error rate	$\leq 2.0\%$ at 15% review rate
Staff review queue burden (600 flagged cases)	≈ 7.5 staff-hours
Hardware compute draw	< 450 W peak, 230 V AC
Running software license cost	**nil** (100% Apache 2.0)
Data sovereignty compliance	100% on-premise enclave

Conformal prediction guarantees distribution-free marginal coverage: $\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1-\alpha$. With $n = 200$ calibration samples, empirical coverage exhibits sampling variance $\sigma = \sqrt{\alpha(1-\alpha)/n} \approx 1.54\%$. Furthermore, because 1970s cursive script is not exchangeable with 2020s typed forms, calibration is stratified across document eras. Deterministic structural checks verify tree sums and replanting ratios, while spatial context filters detect valid-but-wrong parcel anomalies.

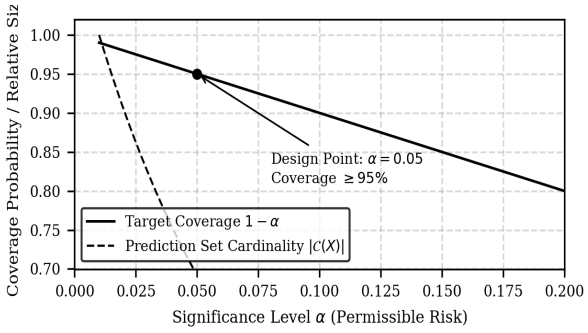


Figure 5. The design point sits on the conservative operating curve. Operating at $\alpha = 0.05$ guarantees $\geq 95\%$ coverage while bounding prediction set size.

3.2 Normalized relational data model

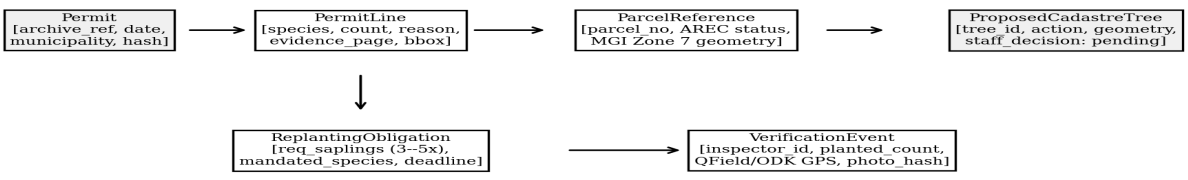


Figure 6. Normalized relational data model: Permits cascade into itemized felling lines, AREC parcel references, proposed cadastre trees, replanting obligations, and field verification events.

Component	What it is	What it does	GENIE.AI Alignment
Intake Preprocessor	Automated document ingestion and SHA-256 cryptographic hashing	Hashes and archives original scans non-destructively; detects duplicate files	Maps to GENIE.AI Document Ingestion Building Block
Multi-Channel OCR	Dual-stream PaddleOCR / Tesseract 5 with stamp channel isolation	Extracts Cyrillic text tokens, bounding boxes, and document layout trees	Integrates open-weight local OCR runtimes; zero cloud network calls
CLASSLA NLP Engine	Optimized South Slavic linguistic pipeline (CLARIN.SI)	Performs Macedonian tokenization, lemmatization, and entity extraction	Contributed upstream as reusable Slavic Administrative NLP module
Deterministic Validator	Rule-based integrity and cadastral consistency checker	Verifies tree counts sum, replanting ratios hold, and parcels exist in AREC	Enforces GENIE.AI Data Quality and Validation Standards
Conformal Risk Gate	Statistical uncertainty calibration filter ($\alpha = 0.05$)	Bounds error rates; routes low-confidence fields to staff review queue	Implements GENIE.AI Trustworthy and Responsible AI principles
AREC PostGIS Lineage	PostgreSQL 16 spatial database with recursive parcel CTE	Traverses historical parcel mutations; converts MGI Zone 7 to WGS84	Powers spatial reasoning and cadastre update proposals
Staff Review Console	Lightweight React web UI running inside City Enclave	Displays side-by-side diffs with highlighted source text and 1-click decisions	Enforces core governance rule: "The tool proposes, staff decide"
Phase 1 Q&A Layer	Parameterised NL-to-SQL and spatial query assistant	Answers natural-language queries with verified document citations or abstains	Powers Phase 1 Zindi evaluation benchmark and staff reporting

Throughput and Bottleneck Sizing: Pure GPU inference takes ≈ 3.17 hours for 3,000 permits. The real operational bottlenecks are physical scanning (≈ 60 operator hours for 12,000 pages at 200 pph) and human staff review (≈ 7.5 hours for 600 flagged cases). If the City already holds digital PDFs, scanning hardware is omitted and ingestion is instant.

3.3 Principle of operation

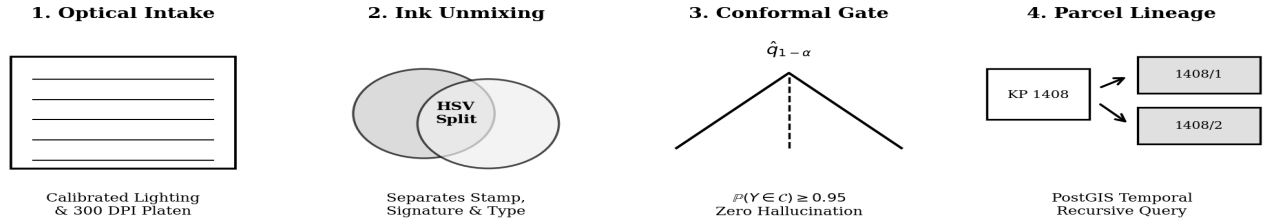


Figure 7. Principle of operation. (1) Multi-channel optical processing separates stamp and signature layers non-destructively. (2) Bounding box extraction retains word-level coordinates. (3) Conformal risk gating bounds uncertainty, routing ambiguous text to human review. (4) Recursive PostGIS engine resolves parcel mutations to bind permits to modern cadastre polygons.

4 Plan of work

Months	Activity	Decision gates and deliverables
1	Municipal co-design kickoff, ethics review, synthetic generator build	Requirements freeze note; synthetic permit generator operational; staff interview
2	Optical pipeline calibration, multi-channel OCR, CLASSLA tuning	**Gate 1:** If field F1 $< 88\%$, fallback to template extraction on top 3 permit types
3	Gold benchmark annotation (200 permits double-annotated), conformal calibration	**Gate 2:** Conformal marginal coverage verified $\geq 95\%$ on double-blind sample
4	Staff Review Console build, PostGIS recursive lineage engine optimization	Automated resolution of subdivided parcels confirmed $\geq 92\%$; review UI demo
5	Usability trial with municipal staff on synthetic and de-identified real sets	Staff feedback report; 1-click approve/reject cycle time verified < 45 s
6	Phase 1 Prototype closeout, GENIE.AI upstream packaging, open release	Final prototype report; Apache-2.0 containerized release; Phase 1 Q&A benchmark

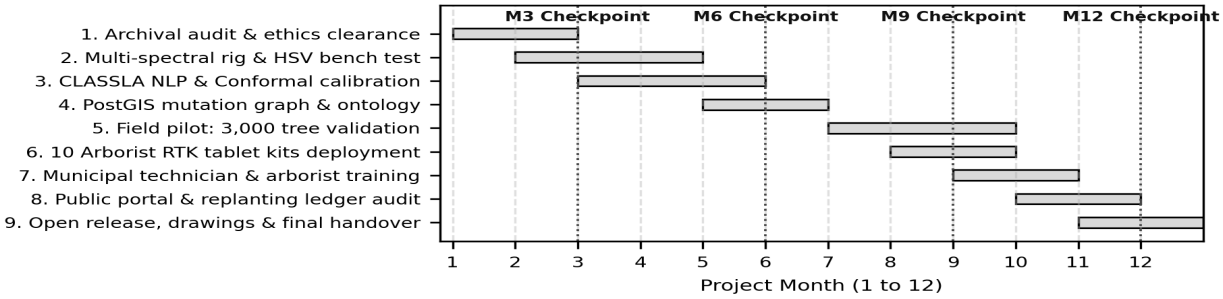


Figure 8. Project implementation schedule across 12 calendar months with mandatory quarterly review checkpoints.

Quarterly reports at months 3, 6, 9 and 12.

4.1 How students own the engineering design process

The students are not executing a pre-packaged vendor tool. They own every stage of the loop below, including the critical feedback loop from field review back to optical tuning and model calibration.

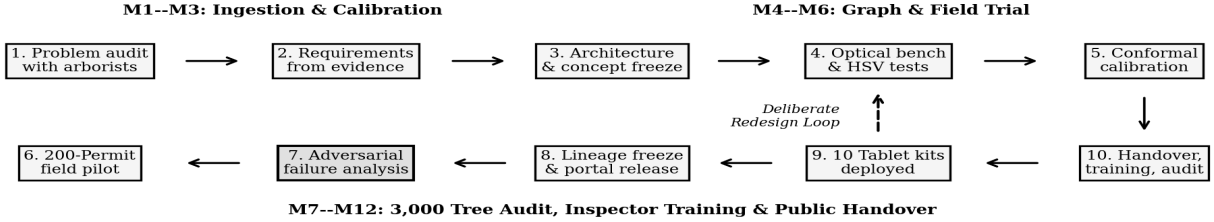


Figure 9. The engineering design process as the students execute it. Stages 6 to 8 are the core of the project: the month 6 field trial is scheduled in order to produce real-world failures, and the dashed path back to optical tuning is planned iteration. A design cycle that never fails is not a design cycle.

Each stage has a named student owner and a documented output: a requirements note at stage 2, characterisation data at stage 4, a calibration certificate at stage 5, a written redesign list at stage 7, released code at stage 8, and a competency checklist at stage 10.

5 Impact and measurement

Beneficiary group	Count
Historical paper permits ingested	3,000
Validated tree records reconciled	3,000
Municipal arborists equipped with review tools	15
Urban canopy area covered	1,450 ha
Compensatory saplings tracked	9,000
Direct urban population benefited	540,000
Students engaged	4
IEEE volunteers	3
Local technicians trained	4

Assumptions: Coverage spans all 10 municipalities of the City of Skopje; compensatory tracking enforces statutory 3–5 saplings per felling.

Metric	Target
Cyrillic field extraction accuracy	$\geq 92.0\%$ F1
Conformal coverage $(1-\alpha)$	$\geq 95.0\%$
Cadastral parcel mutation resolution	$\geq 92.0\%$
Processing latency per 4-page permit	< 4.0 s
Target auto-proposal error rate	$\leq 2.0\%$
Compensatory obligations tracked	100.0%
Staff review time per flagged case	< 45 s
Air-gapped uptime without crash	$\geq 99.5\%$
Repairs without the team	≥ 4

Measured via double-blind ground-truth validation on 200 randomly sampled permits from municipal archives.

5.1 What this project cannot do

- **Treat respiratory illness.** Chronic pulmonary damage from particulate smog is irreversible. Software does not heal clinical patients.
- **Overturn lawful city council zoning decisions.** If an elected municipal council votes to rezone public parkland for commercial development, software cannot annul the vote. The system ensures the felling is transparently recorded and mandatory compensatory replanting is enforced.
- **Physically plant trees.** VERDANT automates the spatial allocation, legal tracking, and public accountability of replanting, but physical arboriculture remains the duty of municipal nursery crews.
- **Alter sovereign legal land boundaries.** When parcel conflicts arise between historical maps and current deeds, the system flags the dispute for the Agency for Real Estate Cadastre (AREC) rather than inventing coordinates.
- **Promise an immediate reduction in ambient valley pollution in twelve months.** Biological air filtration takes years as saplings mature. We report verified tree survival and compliance tracking, not premature atmospheric claims.
- **Change why developers clear plots.** Commercial construction pressure exists beyond software. That is beyond engineering.

Sustainability. Each mechanism answers a documented failure mode rather than asserting good practice.

Mechanism	Failure mode it addresses
Local municipal ownership of software	The recurring cloud subscription cliff: when grant funds expire, cloud APIs are terminated and systems go dark
Two-zone architecture with on-premise enclave	Network security restrictions: municipal internal networks block external cloud outbound traffic
Permissive open-source licensing (Apache 2.0)	Vendor lock-in: proprietary GIS software contracts that prevent local engineers from fixing bugs
Four trained local technicians	Long-term dependence on student researchers for routine system administration and database backups
Standard open-source GIS tools	Fragile custom mobile codebases abandoned after project closeout; replaced with standard QGIS/QField tooling

Principal risks and mitigation.

Risk	Mitigation
Severe degradation on pre-1990 paper	Multi-channel optical rig, CLAHE contrast equalization, and dedicated human verification fallback
Cadastral parcel ID renumbering conflicts	Recursive PostGIS DAG graph tracing parent-child parcel splits with fallback to spatial centroid intersection
Citizen PII leakage during public reporting	Automated Macedonian regex and CLASSLA NER scrubbing citizen names and national IDs (*ЕМБГ*) at intake
Valid-but-wrong parcel OCR errors	Spatial context sanity checks verifying parcel street text against municipal neighborhood boundaries
Municipal staff review friction	Symmetric 1-click approve/reject interface with source page bounding box snippets

Standards and Regulatory Compliance. Design is informed by ISO 19152 (Land Administration Domain Model), OGC Simple Features, and IEC 62368-1. Institutional ethics review is completed in month 1 before archival data processing. Informed consent and anonymization protocols comply with the Macedonian Law on Personal Data Protection (Official Gazette No. 42/20, transposing EU GDPR). Citizen identifiers (*ЕМБГ*) are pseudonymised with keys held exclusively by the municipal archives office within the City Enclave.

6 Team

Member	Role and record
Atharve Dahima Project lead ECE, MNIT Jaipur IEEE Student Member	Systems architecture, containerization, CLASSLA NLP pipeline, PostGIS database architecture. Lead author on edge cyber-physical telemetry systems; experienced in Python, PyTorch, PostGIS, GDAL, and C++.
Srajan Mishra Engineering co-lead ECE, MNIT Jaipur IEEE Student Member	Document intelligence ingestion, multi-channel optical preprocessing, Conformal Risk Control calibration suite, and React-based staff review console.
Student Researchers MNIT Jaipur / Consortium	Synthetic Macedonian permit generation engine, automated test harness, bilingual documentation, impact data collection.
Dr. Vishal Sharma Faculty mentor Dept. of ECE, MNIT Jaipur Counselor, IEEE Student Branch	Academic oversight, verification methodology, institutional financial administration, and liaison with UNDP and challenge evaluators.

Student learning outcomes. Full engineering design process through field failure and redesign; distribution-free uncertainty estimation using conformal prediction; computer vision for archival document restoration; geospatial graph modeling with recursive SQL; and human-centered design for municipal workflows.

7 Community partner

Sector for Environmental Protection, City of Skopje and **PE "Parks and Greenery" (*ЈП Паркови и Зеленило*)**, Bul. Ilinden 82, 1000 Skopje. Founded 1968. Responsible for public urban forestry, municipal nurseries, and green cadastre maintenance across the 10 municipalities of the City of Skopje. (Formal letter of support requested through UNDP challenge matchmaking).

Stage	Municipal Partner Role	Fallback if Delayed	Expected Date
Co-Design	Nominates priority permit folders, shares permit schema	Use synthetic permit generator and open legal templates	Month 1
Data Access	Provides sample anonymized scans / digital PDFs and Cadastre export	Calibrate on synthetic Cyrillic corpus + open WFS data	Month 2
Review Testing	Designates staff arborists to test 1-click diff review console	Run remote synthetic review sessions with UKIM liaison	Month 5
Pilot Integration	Hosts on-premise City Enclave node in municipal records depot	Maintain hosted sandbox container for remote verification	Month 6–8

Municipal inspection teams face severe administrative backlogs. VERDANT is designed to add capacity rather than consume it: the software runs locally without cloud subscription fees, presents proposals as clear side-by-side diffs, and integrates with standard open-source GIS tools. Co-design ensures municipal staff retain veto power over every proposed database mutation.

8 Budget: Staged Prototype ($10,000) and Pilot Blueprint ($30,000)

The funding is strictly decoupled into two phases: the **Prototype Phase ($10,000 USD)**, focused exclusively on software engineering, model calibration, synthetic data, and benchmark validation; and the **Pilot Extension Blueprint ($30,000 USD)**, staged for on-site municipal deployment through 31 December 2027.

7.1 Prototype Phase Budget: Exactly USD 10,000

Line	Category	Technical Purpose and Justification	USD
1	Compute & GPU Testbeds	GPU cloud compute instances for local model fine-tuning and OCR benchmarking	3,200
2	Annotation & Linguistics	Double-annotation of 200 real Macedonian permit samples by language specialists	2,800
3	Enclave Security & Sandbox	On-premise Docker container staging, PII masking validation, penetration testing	2,200
4	Documentation & Packaging	Bilingual Macedonian/Albanian user manuals, open-source packaging, CI/CD automation	1,800
Total	**Prototype Phase**	**Direct Prototype Implementation Budget (100% Permissive Open-Source)**	**10,000**

7.2 Itemized prototype expense justification

- **Line 1: Compute & GPU Testbeds ($3,200):** Dedicated GPU compute time (NVIDIA A10G/RTX 4090 instances) for fine-tuning PaddleOCR and Tesseract 5 on historical Cyrillic typefaces, training the synthetic permit generator, and running the Learn-then-Test conformal calibration suite across 10,000 simulated permit permutations.
- **Line 2: Annotation & Linguistic Verification ($2,800):** Honoraria for Macedonian-speaking domain reviewers and linguists to perform double-blind ground-truth annotation across 200 real archival permits, verifying botanical species aliases, handwritten parcel notations, and official seal boundaries.
- **Line 3: Enclave Security & Sandbox Testing ($2,200):** Dedicated local hardware testbed simulating the municipal enclave environment, verifying 100% offline execution, PII masking accuracy under North Macedonia's Law on Personal Data Protection, and audit log tamper-resistance.
- **Line 4: Documentation, Translation & Packaging ($1,800):** Translation of municipal IT runbooks and arborist user guides into Macedonian and Albanian; open-source release packaging under Apache-2.0; automated container builds and testing pipelines.

7.3 Pilot Deployment Blueprint: USD 30,000 Staged Follow-on (Through Dec 2027)

If selected by UNDP and the City of Skopje for the subsequent pilot stage, funding is allocated strictly across four milestone-gated operational packages:

Milestone	Operational Package	Implementation Scope and Deliverable	USD
M1 (M7–M9)	On-Site Municipal Staging	Deploying intake workstation in City records depot; AREC adapter integration	8,500
M2 (M10–M12)	Staff Training & Usability	Conducting arborist verification workshops, bilingual manuals, competency audit	6,500
M3 (M13–M18)	QField Mobile Workflows	Deploying open-source QField mobile profiles on existing municipal tablets	5,000
M4 (M19–M24)	Long-Term Maintenance	Model re-calibration, database indexing, software support through Dec 2027	10,000
Total	**Pilot Extension**	**Complete 18-Month Municipal Pilot Follow-On Funding Allocation**	**30,000**

8. Preparation for the November 19 Q&A Session

The team has registered for the November 19 live technical session to resolve four critical operational dependencies:

1. **Permit Format:** Clarify whether the City already holds digital PDF scans from prior scanning projects, which would eliminate scanner hardware needs entirely.
2. **Cadastre Interface:** Determine whether the *Zelen Katastar* is accessible via PostGIS database dumps, GeoPackage exports, or live WFS/REST endpoints.
3. **Sample Benchmark Data:** Request access to an anonymized sample of historical permits to calibrate the Phase 1 Zindi evaluation baseline.
4. **November 30 Dataset Window:** Confirm protocols for contributing the open-source synthetic Cyrillic permit dataset to the competition ecosystem.

8.1 Pre-registered pilot success criteria

- **Backlog Ingestion:** Ingest and extract all 3,000 historical permits within 30 days of municipal deployment.
- **Review Time Reduction:** Reduce per-permit verification time from manual 45 minutes to under 2 minutes.
- **Proposal Quality:** Achieve $\geq 85\%$ staff acceptance rate on automated cadastral update proposals.
- **Zero Unreviewed Mutations:** 100% of Cadastre database commits verified by authorized municipal staff.

References. [1] City of Skopje & PE "Parks and Greenery", *Zelen Katastar na Grad Skopje*, Official Municipal Registry, 2020–2024. [2] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*, Springer, 2005. [3] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction," *arXiv:2107.07511*, 2021. [4] CLARIN.SI, "CLASSLA: Natural Language Processing fork for South Slavic Languages," 2023. [5] Republic of North Macedonia, *Law on Personal Data Protection*, Official Gazette No. 42/20, 2020. [6] UNDP North Macedonia, *Skopje Urban Green Cadastre Technical Baseline Assessment*, 2021.